INCH-POUND
MIL-DTL-24643/76
22 April 2014

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -20 °C TO +105 °C, 5000 VOLTS, 133 PERCENT RATED, TYPE LS5KVTEPSG

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

The following commercial specifications shall be used in conjunction with this military specification to the extent indicated herein: UL 1072, ANSI/NEMA WC 74/ICEA S-93-639, ANSI/NEMA WC 53/ICEA T-27-581, ICEA T-24-380, ICEA T-32-645, and IEEE 1580. Ethylene propylene rubber (EPR) consisting of EPM (a copolymer of ethylene and propylene and ASTM D1418-class rubber) or EPDM (a terpolymer of ethylene, propylene, and a diene third monomer component used for cross-linking) shall be used. All EPR materials shall be cross-linked. Sulfur curing is not allowed. A peroxide derivative is the preferred curing method. XLPE is not allowed. Only extruded thermoset materials shall be used for this cable design.

Construction, Watertight with Circuit Integrity

Component 1 (three shielded conductors):

First	-	Copper-stranded conductor, coated or uncoated, with extruded or pumped semi-conductive waterblocking material (see table I for size).
Second	-	An optional semi-conducting separator tape, applied helically with a minimum of 25 percent overlap, may be used between the stranded conductor and the extruded thermoset semi-conducting conductor shield.
Third	-	Extruded thermoset semi-conducting conductor shield, minimum wall thickness 0.006 inch (in accordance with ANSI/NEMA WC 74/ICEA S-93-639).
Fourth	-	Uniformly extruded thermoset EPR insulation. Insulation Level 133 percent. Minimum thickness of 0.085 inch. Maximum thickness of 0.120 inch.
Fifth	-	Extruded semi-conducting insulation shield, minimum wall thickness 0.024 inch (in accordance with the table for Thicknesses of Extruded Conductive Nonmetallic Covering, column C, of UL 1072).
Sixth	-	An optional semi-conducting separator tape may be used between the extruded semi-conducting insulation shield and the tin-coated copper braided wire shield.
Seventh	-	A semi-conducting warning shall be applied in accordance with the section for Identification of Conductive Nonmetallic Covering Portion of Insulation Shielding of UL 1072.
Eighth	-	Tin-coated copper braided wire shield, 95 percent minimum coverage, no wire size less than 34 AWG allowed for ground fault protection. Shield voids filled with extruded or pumped semi-conductive waterblocking material.

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Ninth - An optional binder.

Tenth - Extruded XLPO jacket, 45 mils (0.045 inch) minimum average.

NOTE: Identification of phase conductors on their extruded jackets shall be in accordance with 3.4.8.2 of MIL-DTL-24643 identification method 1 or identification method 3, or identification method 5 modified as follows: the underlined letters "A", "B", and "C" shall be printed on the component extruded jackets. The letters shall be approximately 3/16 inch high with alternate legends inverted, and shall be printed at intervals not exceeding 1½ inches. All three-phase conductors shall be printed. The letter axis may be perpendicular or parallel to the longitudinal axis of the component, and shall be printed in a contrasting color: preferably white ink on black or dark background, or black ink on white or light background.

Component 2 (three insulated ground wires):

First - Copper conductor, coated or uncoated, 19 strands minimum (see table I for size).

Second - Thermoset insulation, extruded. Minimum average wall thickness of 0.040 inch.

Third - Optional covering.

Assembly:

First - The three shielded conductors shall be cabled together with a lay in accordance with 3.4.4 of MIL-DTL-24643. One ground wire shall be placed in each interstice. Fillers may be used as necessary to form a firm, well-rounded assembly. Blocking compound shall be applied to provide waterblocking (blocking of all voids between the conductor assemblies, the filler materials, and the overall binder or shield). Filler material shall be compatible with other cable components. Ground wires only must be color coded green in the insulation, and in the optional covering, if applied. (If optional covering is not green, green color must be visible through the applied optional covering.) Ground wires may also be identified by printed identification in green ink on the outer surface of the ground wires.

Second - An optional binder.

Third - Braided shield of tin-coated copper, 95 percent minimum coverage, no wire size less than 34 AWG allowed for ground fault protection. Shield voids filled with extruded or pumped non-conductive waterblocking material.

Fourth - An optional binder.

Fifth - Extruded XLPO jacket. Minimum average wall thickness of 0.125 inch.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

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TABLE I. Cable details.

Military part no. M24643/76	Type and size	Conductor size/ ground sizes		Cable overall diameter max. (inch)	Conductor resistance (ohms)	Insulation resistance (megohms)	Accelerated service loading (amperes)
		Navy STD	MCM				
-01UO	LS5KVTEPSG -250	250	250MCM	2.900	0.0453 / Gnds: 0.3330	900	535
-02UO	LS5KVTEPSG -400	400 (127)	400MCM	3.300	0.0273 / Gnds: 0.3330	750	750

REQUIREMENTS:

Qualification required.

INSPECTION:

Partial discharge (PD) level (corona) test: The finished cable shall be measured for partial discharge extinction level in accordance with ICEA T-24-380. The test shall not be performed on extruded cable until a minimum of seven days has elapsed following the insulation extrusion process. The inception voltage shall be equal to the voltage at the minimum acceptable partial discharge extinction level specified below. A Hipotronics partial discharge detector model DDX 7000, or equal, shall be utilized.

Acceptance criteria: During each test, there shall be no evidence of arcing or corona discharge within the insulation system of each conductor. The extinction level, when defined as the voltage at which the partial discharge is less than 5 picocoulombs, shall be 18 kV. In the event that the inception is not reached and the 18 kV is attained, the test shall be terminated. In such a case, the cable shall be considered partial discharge resistant and as having passed the test.

Bending radius (permanent training) (min.): Shall be 8 times the cable maximum specified overall diameter (OD). The term “minimum bend radius” indicates the radius of the smallest diameter circle around which the cable may be bent, resulting in the maximum amount of cable bend allowed. This bend radius applies to cable being bent in place, with no tension applied to the cable or conductors.

Dimensional measurements: This test shall be performed optically in accordance with ANSI/NEMA WC 53/ICEA T-27-581.

Basic Electricals:

Conductor resistance (ohms/1000 feet at 25 °C, max.)	See table I
Voltage withstand (volts, root mean square, min.)	
Conductor to shield	18000
Ground wire to all shields	2000
Component shield to all other shields	2000
Insulation resistance (megohms/1000 feet, min.)	
Conductor to shield	See table I
Ground wire to all shields	225

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Component shield to all other shields	100
Conductor, ground, and shield continuity	No failure
Group tests for partial discharge:	
1. Partial discharge, conductor to shield, maximum at 18 kV AC (See above for test procedure/information.)	5 picocoulombs
2. AC withstand, conductor to shield, maximum at 18 kV AC	18kV (volts, root mean square, min.)
Jacket flaws	No failure
Group A:	
Real time on-line cable production monitoring shall be utilized to ensure that tolerance and thickness requirements are met.	
Watertightness (see MIL-DTL-24643 for limits of water leakage)	No failure
Group B:	
Adhesion and stripability of extruded layers	This test shall be performed in accordance with ANSI/NEMA WC 53/ICEA T-27-581.
Cold bending cable	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Outer cable jacket (when tested at 200 °C)	50
Drip (95±1 °C)	Zero
Physicals (unaged)	
Insulation (extruded)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Outer cable jacket	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Group C:	
Physicals (aged) air oven	
Insulation (extruded)	
Tensile strength (lb/in^2, min.)	75
Elongation (percent, min.)	75

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Outer cable jacket	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (outer cable jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Cable fill (sealant) removability	No failure
Shrinkage	No failure
Component and overall shield (conformance to material, construction and coverage)	No failure
Surface transfer impedance	
Milliohms per meter (max.)	Not applicable
EMP response (dB, min.)	Not applicable

Group D:

Flame propagation (cable)	No failure
Gas flame (1 hour)	No failure

Qualification Inspection:

Qualification inspection shall include basic electricals; Groups A, B, C, and D; plus the following:

Salt water immersion: This test shall determine the ability of the completed cable to withstand the effects of salt water and sea spray that may occur during marine service. The Salt Water Immersion Test detailed in IEEE 1580 shall be performed. Specimen test length shall be 5.25 feet (1.60 meters) in lieu of the 3.6 feet (1.1 meters) specified.

Acceptance criteria: After immersion in salt water, the cable shall comply to the following:

a. The cable shall pass the voltage withstand test of MIL-DTL-24643.
b. There shall be no degradation of the mechanical properties of the jacket or insulation such that they would crack when wound around a mandrel having a diameter equal to sixteen times the sample maximum specified OD.
c. The insulation and jacket shall not degrade to the point where either will crack or separate from the cable during the conditioning or during the testing required in items (a) and (b) above.

Compatibility of semi-conducting materials	This test shall be performed in accordance with ICEA T-32-645.
Wafer boil	This test shall be performed in accordance with ANSI/NEMA WC 53/ICEA T-27-581.
Cold working (minus 20±2 °C)	No damage
Gas flame (3 hours)	No failure
Accelerated service (see table I for load current)	No failure
Aging and compatibility (cable) (125±5 °C)	No failure

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Abrasion resistance (outer cable jacket) (scrapes, min.)	75
Acid gas equivalent (cable) (percent, max.)	
Jackets (all)	2
Fillers	2
Insulation	18
Semi-conducting materials	18
Halogen content (cable) (percent, max.)	
Jackets (all)	0.2
Fillers	0.2
Insulation	0.2
Semi-conducting materials	0.2
Immersion (outer cable jacket)	
Elongation (percent of unaged, min.)	50
Tensile strength (percent of unaged, min.)	50
Smoke index (cable) (max.)	
Jackets (all)	25
Fillers	45
Insulation	45
Semi-conducting materials	45
Toxicity index (cable) (max.)	
Jackets (all)	5
Fillers	5
Insulation	1.5
Semi-conducting materials (Not applicable to semi-conductive tape.)	1.5
Durometer (outer cable jacket) - Type A (hardness, min.)	80
Weathering (outer cable jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LS5KVTEPSG all sizes	1000

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Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2014-031)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – CC

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